DBMS PROJECT

Smart Hotel Booking System

A DBMS project combining hotel booking management, cancellation prediction, and dynamic pricing.

PYTHON • STREAMLIT • SQLITE • PANDAS • NUMPY • SCIKIT-LEARN

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THE OPERATING RISK

Manual booking creates avoidable operational risk

Availability is hard to track.
Paper registers and spreadsheets can allow double bookings.

Repeated data creates errors.
Hand-calculated bills and updates become difficult to maintain.

The system centralizes decisions.
It checks date overlaps, predicts cancellation risk, and suggests room prices.



The risk begins when availability, billing, and decisions live in separate places.

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SYSTEM ARCHITECTURE

Three layers turn a booking into a decision

1

Streamlit collects guest, room, date, and payment inputs.

2

Python validates availability, calculates bills, runs prediction and pricing logic, and sends results to the interface.

3

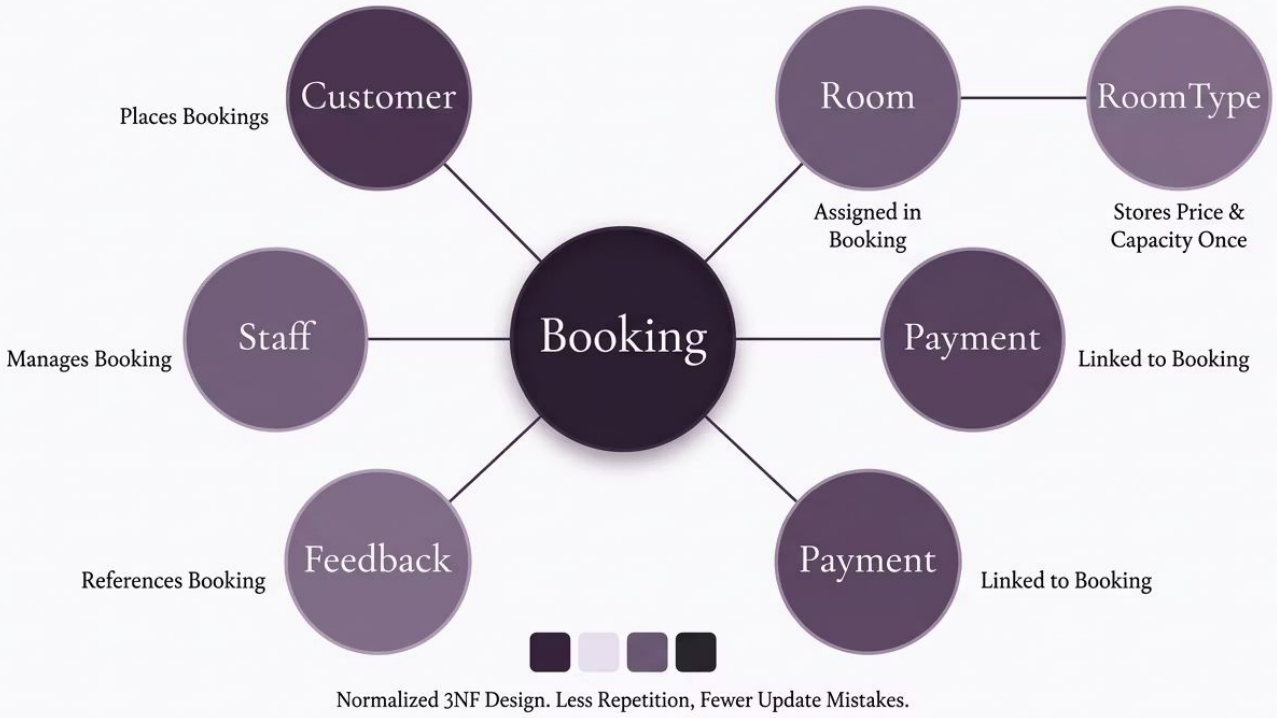
SQLite stores booking data and enforces relational rules through keys, constraints, triggers, and views.

User → Streamlit → Python → SQLite / prediction logic → results on screen

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DATA MODEL

Seven linked tables keep hotel data consistent



Central Booking table · primary keys · foreign keys · normalized 3NF design

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DBMS AUTOMATION

DBMS automation protects the room lifecycle

The database turns rules into repeatable actions.

Booking

Check-In



Reporting & Billing

Check-Out / Cancel

Billing follows nights × room price × 1.12 GST.

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END-TO-END WORKFLOW

From booking form to dashboard insight

01

Enter

Guest details, dates, room type, and deposit.

02

Check

Availability and estimated bill.

03

Confirm

Save the booking and payment details.

04

Manage

Check in or check out and see room status update.

05

Review

Cancellation prediction, dynamic price suggestion, and SQL-based reports.

Core modules: booking management, prediction, pricing, database, and dashboard.

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EVALUATION AND NEXT STEPS

Practical today, extensible tomorrow

Advantages

Easy web interface, centralized records, local storage, automated billing, and decision support.

Current limits

Sample data, single-user SQLite, no login or online payment, and a basic interface.

Future scope

Role-based access, email/SMS notifications, online payments, stronger real-world models, advanced demand pricing, and MySQL or PostgreSQL for multi-user scale.



One low-cost system brings database integrity, automation, and smarter hotel operations together.

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